DSCI 550 – Project 2 Proposal

Part 1. Group Information

Member	Email
Sahana Nadig	snadig@usc.edu
Xinchen Geng	gengx@usc.edu
Zheyu Chen	zheyuc@usc.edu
Zexu Gao	zexugao@usc.edu

Part 2. Project Idea and Competition

We propose participating in Numerai Classic (https://numer.ai/), an ongoing machine learning tournament where participants predict and rank future relative stock performance using anonymized financial data. We will use the Classic v6.0 dataset, released on October 16, 2026, which is freely available through the numerapi Python package. Each record represents a stock on a specific date, with features and targets measuring future performance over 60 business days. New live data is released Tuesday through Saturday, with v6 submissions beginning November 6, 2026. Unlike traditional competitions, Numerai updates provisional leaderboard scores as real market results become available, while final scores take longer to settle.

Item	Description
Row	One stock (numerai_ticker) on one date
Features	Anonymized numeric values describing each stock, listed in features.json
Main target	future relative performance over 60 business days
Live data	live.parquet, released each round, Tuesday to Saturday

Our research question is: Do changes in feature selection and model design improve out-of-sample Numerai CORR consistently across historical market periods, and are those improvements visible in early live scoring? Financial outcomes are noisy and evolve over time, so models that perform well on past data may fail on later dates. This makes robust validation, not simply obtaining the highest one-time score, an important data science problem.

Public-solution check (October 26, 2026): We reviewed Numerai's competition documentation and leaderboard, the Unity release discussion on the Numerai forum, its public example-scripts on GitHub, and an independent public end-to-end training/submission pipeline. Reproducible starter models and workflows do exist; however, they cannot reveal future market targets or permanently solve an ongoing forecasting problem. We will conduct our own controlled experiments and document original changes.

Part 3. Proposed Solution (Methodology)

We will build on Numerai's official LightGBM example model using standard Python libraries, including pandas, LightGBM, and numerapi. We will change one component at a time to understand how each modification affects model performance.

1. **Baseline:** We will reproduce Numerai's example LightGBM model using the small feature set as our baseline.
2. **Validation:** We will preprocess the historical data and split it chronologically, training on earlier dates and validating on later ones. We will leave a gap of at least 60 business days between training and validation to reduce data leakage from overlapping target periods. Rows without available target values will be excluded from training.

3. **Experiments:** We will compare small, medium, and full feature sets, experiment with additional target columns, and test whether combining multiple models improves prediction performance. We will change one factor at a time to evaluate its effect.
4. **Live Submissions:** We will submit the baseline starting November 6 and add our best experimental model once it is ready. Both models will use the same live data in subsequent rounds and submit predictions through an automated Python script.

Part 4. Proposed Evaluation

Our primary competition-aligned measure is Numerai CORR (rank-normalized correlation with the target), not ordinary classification accuracy or raw Pearson correlation. We will also report mean and standard deviation of CORR across dates, the paired difference between models on the same validation dates, model runtime, and MMC where available. We will compare the official-inspired LightGBM baseline against each modification, show date-by-date score trends, and summarize failure periods. Live CORR will only provide early external evidence: scores update as market information matures, while only resolved rounds contribute to final leaderboard rankings. Therefore, the historical holdout will support our main conclusions, and provisional live scores will be described as preliminary rather than final results.

The official model leaderboard lists valois200_1024 first, with a displayed return of +126.8% over one year. Model ranks are based on reputation from resolved historical scores. This snapshot of past performance provides competition context rather than a direct v6 benchmark. Our v6 submissions are planned from November 6, and project rankings will be reported when available, together with their dates and scoring rules.

Part 5. Proposed Timeline and Work Allocation

1. Oct 26 to Nov 5: Set up our Numerai accounts and get the baseline model running.
2. Nov 6: Start submitting the baseline model in every live round.
3. Nov 9 to Nov 20: Run our experiments and pick our best model.
4. Nov 23: Start submitting the second model alongside the baseline.
5. Nov 23 to Dec 8: Keep submitting, collect live scores, and prepare the final report and presentation. The course final exam is Dec 5, so we plan to finish most of the experiments before then.
6. Dec 9 to 14: Final project presentation.

The Numerai tournament runs continuously, and the rounds we submit to will not finish scoring until around February 2027, so no final results or solutions for them will exist while the project is graded. We use Numerai's public example model only as a starting point and compare it against the models we build ourselves.

Individual responsibilities (to be completed by the four group members):

Individual Responsibilities:

- **Sahana Nadig – Data Preparation & EDA:** Download and preprocess the Numerai dataset, analyze feature distributions, and prepare training and validation sets.
- **Xinchen Geng – Baseline & Validation:** Implement the baseline LightGBM model, conduct time-based validation, and calculate historical CORR scores.
- **Zheyu Chen – Model Improvement:** Experiment with different feature sets, target columns, and model combinations to improve prediction performance.
- **Zexu Gao – Live Submissions & Evaluation:** Manage live submissions, track leaderboard scores, and compare baseline and improved model performance.

All four members will collaborate on experiment design, code review, result interpretation, report writing, and the final presentation.

All members will take part in interpretation, code review, report writing, and the presentation.

References

Numerai. Numerai Tournament. https://numer.ai/ (accessed October 26, 2026)
Numerai. Getting Started. Numerai Docs. https://docs.numer.ai/ (accessed October 26, 2026)
Numerai. Submissions. Numerai Docs. https://docs.numer.ai/numerai-tournament/submissions (accessed October 26, 2026)
Numerai. Example Scripts. GitHub. https://github.com/numerai/example-scripts (accessed October 26, 2026)
NumerAPI documentation. https://numerapi.readthedocs.io/ (accessed October 26, 2026)